Unit: 6 Unit 8: Quadratics **Name:** ________________
Topic: Definition of Quadratics and Standard Form **Date:** ____________

Homework (Carolina Reaper)

Q1. A music festival has a $16 \cdot x^2 + 64 \cdot x + 100$ attendance pattern, what is the standard form of this quadratic function?
(A) $16 \cdot (x+2)^2 + 36$
(B) $16 \cdot (x+2)^2 + 4$
(C) $16 \cdot (x+4)^2 + 16$
(D) $16 \cdot (x+2)^2 + 12$
(E) $16 \cdot (x+1)^2 + 25$

Q2. A rhythm pattern is created using $x^2 - 9x + 20$ notes, identify the key features of this quadratic function.
(A) Vertex: (5, -4), Axis of symmetry: x = -1
(B) Vertex: (4.5, -4.25), Axis of symmetry: x = 4.5
(C) Vertex: (4, -4), Axis of symmetry: x = -4
(D) Vertex: (-1, 0), Axis of symmetry: x = -1
(E) Vertex: (4.5, 0), Axis of symmetry: x = -1

Q3. A canvas has dimensions $x^2 - x - 6$ by $x^2 + x + 2$, in standard form what is the area of the canvas?
(A) $x^4 + x^2 - 12$
(B) $x^4 - x^2 - 12$
(C) $x^4 + x^2 + 6$
(D) $x^4 - x^2 + 8$
(E) $x^4 + 2x^2 - 12$

Q4. Which of the following frequencies $x^2 + 5x - 2$ is a quadratic function?
(A) Only A and B
(B) Only C
(C) $x^2 + 5x + 2$
(D) $x^2 - 5x + 2$
(E) $2x^2 + 5x + 1$

Q5. A sound wave has an equation $y = 2x^2 - 5x - 3$, what is the vertex of this quadratic function?
(A) $(-1.25, -4.25)$
(B) $(1.25, 4.25)$
(C) $(-2.5, 4)$
(D) $(0.5, 0)$
(E) $(-0.5, 0)$

Q6. The area of a painting is given by the equation $A = x^2 + 4x + 4$, identify the key features of this quadratic function.
(A) Vertex: (-2, 0), Axis of symmetry: x = 2
(B) Vertex: (-1, 3), Axis of symmetry: x = -1
(C) Vertex: (-2, 4), Axis of symmetry: x = 0
(D) Vertex: (0, 4), Axis of symmetry: x = -2
(E) Vertex: (-2, 0), Axis of symmetry: x = -2

Q7. The length of a rhythm pattern is $x^2 + 2x - 15$, what is the standard form of this quadratic function?
(A) $x^2 + 4x + 1$
(B) $x^2 - 4x + 1$
(C) $x^2 + 2x - 6$
(D) $(x+5)(x-3)$
(E) $x^2 - 3x + 2$

Q8. A painter creates $x^2 + 3x - 9$ unique brush strokes per hour, what is the axis of symmetry of this quadratic function?
(A) $x = -3$
(B) $x = -1.5$
(C) $x = -0.75$
(D) $x = -2.5$
(E) $x = 0$

Open Ended Questions (FRQ)

Q9. Describe the key features of the quadratic function $y = 3x^2 - 2x - 1$, show all work and include a graph.

Q10. Find the equation of the quadratic function in standard form with vertex $(2, 3)$ and axis of symmetry $x = -2$, show all work.

Chef's Tips

- Tip 1: Ensure students use correct units and labels on graphs
- Tip 2: Remind students to simplify expressions and equations
- Tip 3: Encourage students to check their work using different methods